

GRE RELIABILITY IN NS¹ & NNS² ASSESSMENT

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¹NS - Native Speakers

²NNS - Non-Native Speakers

GRE

Link to the test
overview on ETS
website

AGENDA

1. Introduction
2. GRE Test Description
3. RQs
4. Method
5. Results
6. Discussion
7. Implications



INTRODUCTION

Overview

- Study purposes:
(1) identify relationships between two constructs in Verbal Reasoning section of GRE test (hereafter GRE-V);
(2) determining differences and consistency rates of scores in NS and NNS groups
- Language areas: cognitive, metacognitive, vocabulary, and reading
- Context: preparing for graduate school

Literature Review

- Critical reading requires background knowledge (Burton et al., 2009)
- "Critical reading goes beyond reading comprehension: successful students need to do more than decode and form a basic comprehension of what they read" (Burton et al., 2009)
- Critical reading means to "elaborate, reorganize, contrast, integrate, or summarize the given material" (Rohwer & Thomas, 1989, pp. 119, 122)
- GRE program has regarded it appropriate to distinguish analytical ability from verbal and quantitative ability (Powers & Dwyer, 2003)
- Analytical skill is the ability to understand structured sets of relationships, deduce new information from sets of relationships, analyze and evaluate arguments, identify central issues and hypotheses, draw sound inferences, and identify plausible causal explanations (Educational Testing Service, 2000, p. 5)
- Deductive reasoning skills are a part of analytical measure and is not the only ability that "graduate students are called upon to use (Tucker, 1985)
- These kinds of skills are specified by the College Board as needed in order to be prepared for college (Ennis, 1987)

GRE TEST DESCRIPTION

GRE Test Description

- **Norm-referenced test:** 75th percentile means you scored better than 75% of the test takers
- **Placement test:** given to students entering educational institutions to measure their overall knowledge in various subjects
- **Context:** Used for students' entering graduate programs in universities
- **Constructs:** Analytical and Critical Reading skills (definitions are presented on the next few slides)

WHAT DOES GRE MEASURE?

1. Your ability to analyze and evaluate written material and synthesize information obtained from it

2. Your ability to analyze relationships among component parts of sentences

3. Your ability to recognize relationships among words and concepts



Two Item Types

SENTENCE EQUIVALENCE & READING COMPREHENSION



WHAT DO THEY MEASURE?

Sentence Equivalence

- Your ability to reach a conclusion about how a small passage should be completed based on partial information
- It focuses on the meaning of the completed whole.

Construct Type

ANALYTICAL
SKILLS



Reading Comprehension

- understanding paragraphs and larger bodies of text
- distinguishing between minor and major points
- summarizing a passage
- identifying the author's perspective
- analyzing and drawing conclusion from a text
- developing and considering alternative explanations

Construct Type

CRITICAL READING
SKILLS

Major Construct

VERBAL REASONING

“the deployment of linguistic processes according to and in order to satisfy the demands of a reasoning task” (Polk, 1992, p. 64)

Two Subconstructs

Analytical

- identifying and defining problems, setting up strategies to solve them, and monitoring solution process (Powers & Dwyer, 2003)

Critical Reading

- constructing and revising meaning, activating prior knowledge, or monitoring one's comprehension (Pressley & Afferbach, 1995)
- elaborate, reorganize, contrast, integrate, or summarize the given material (Rohwer & Thomas, 1989, pp. 119, 122)

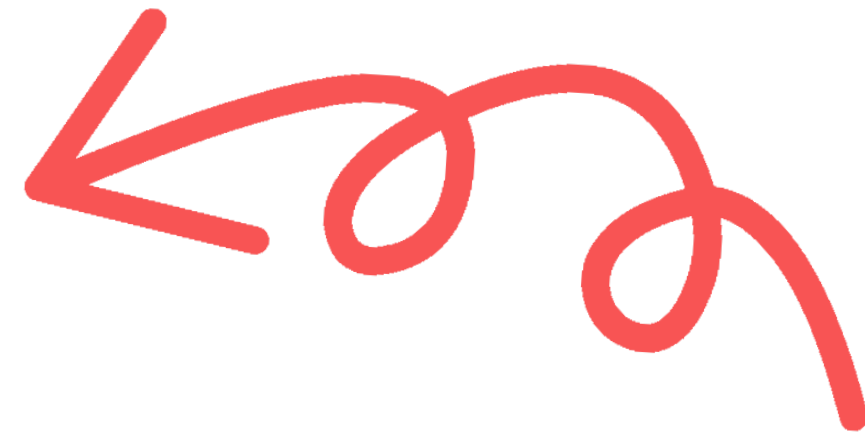
RESEARCH QUESTIONS

CORRELATION

1. What is the relationship between critical reading and analytical skills in GRE-V?



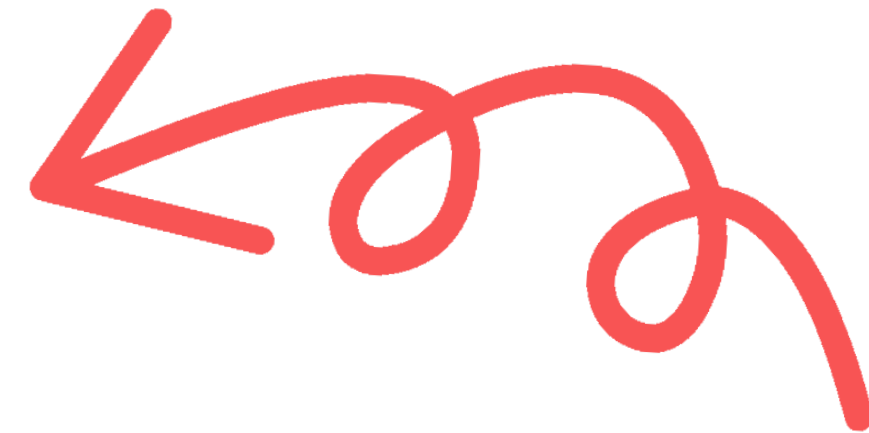
DIFFERENCES



2. To what extent are the test scores of English native speakers and English non-native speakers different?



RELIABILITY



3. How consistent are the test scores
for NS and NNS groups?



METHOD

METHOD

Participants

19 participants

14 female, 5 male

Age of 25 - 34: 78 %

Above 35: 22%

11 NS, 8 NNS

Graduate Students, or at least
have bachelor's degree
earned by the time of taking a
test

Materials

Norm-referenced GRE test

Verbal Reasoning test section

NAU Qualtrics Survey, SPSS

20 questions, 26 items

6 items with multiple answer choices

Multiple-Choice item type

12 Sentence Equivalence items

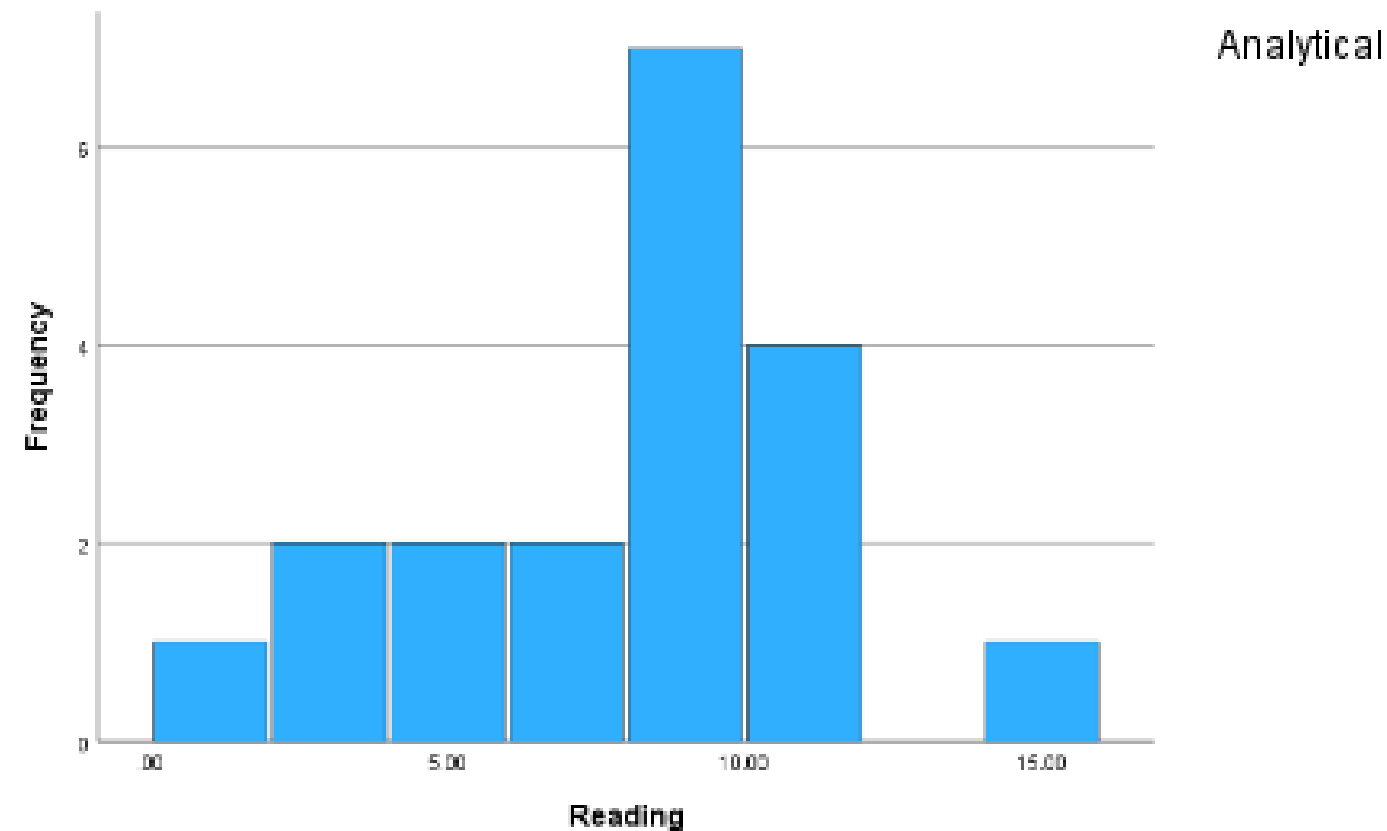
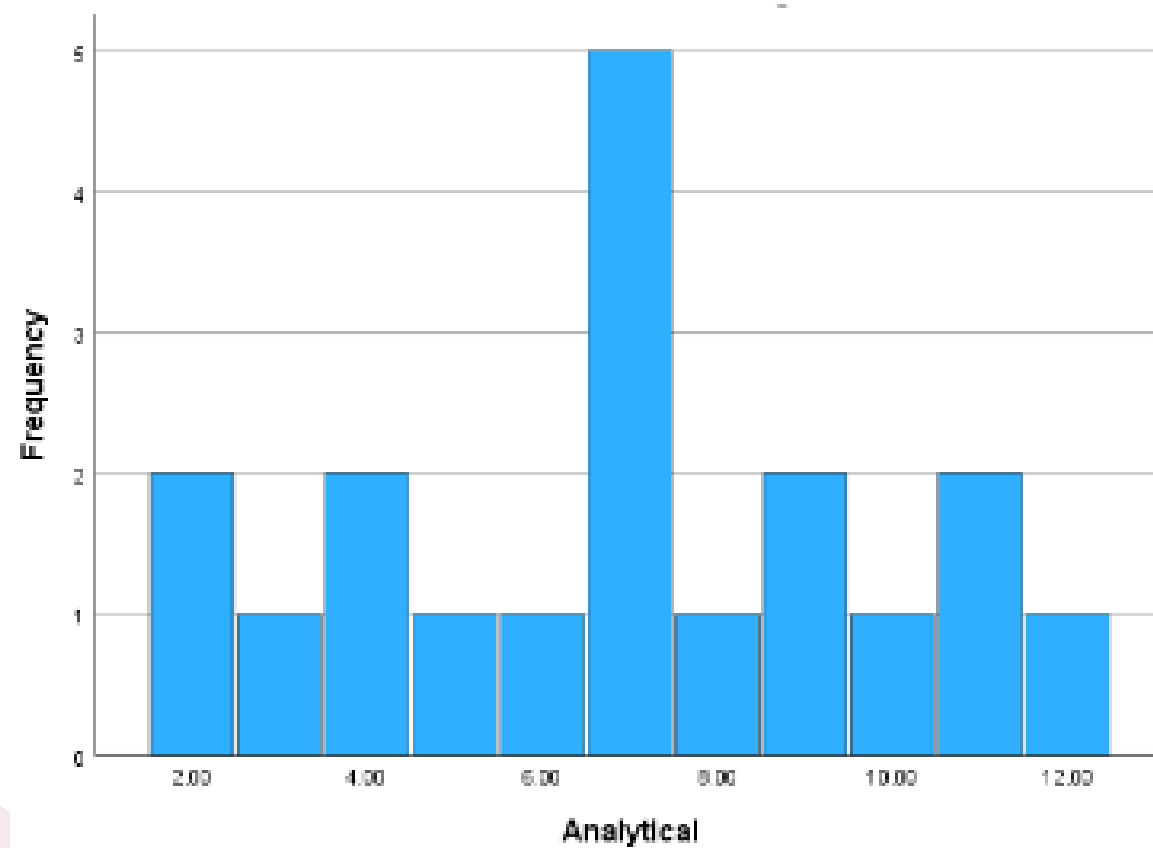
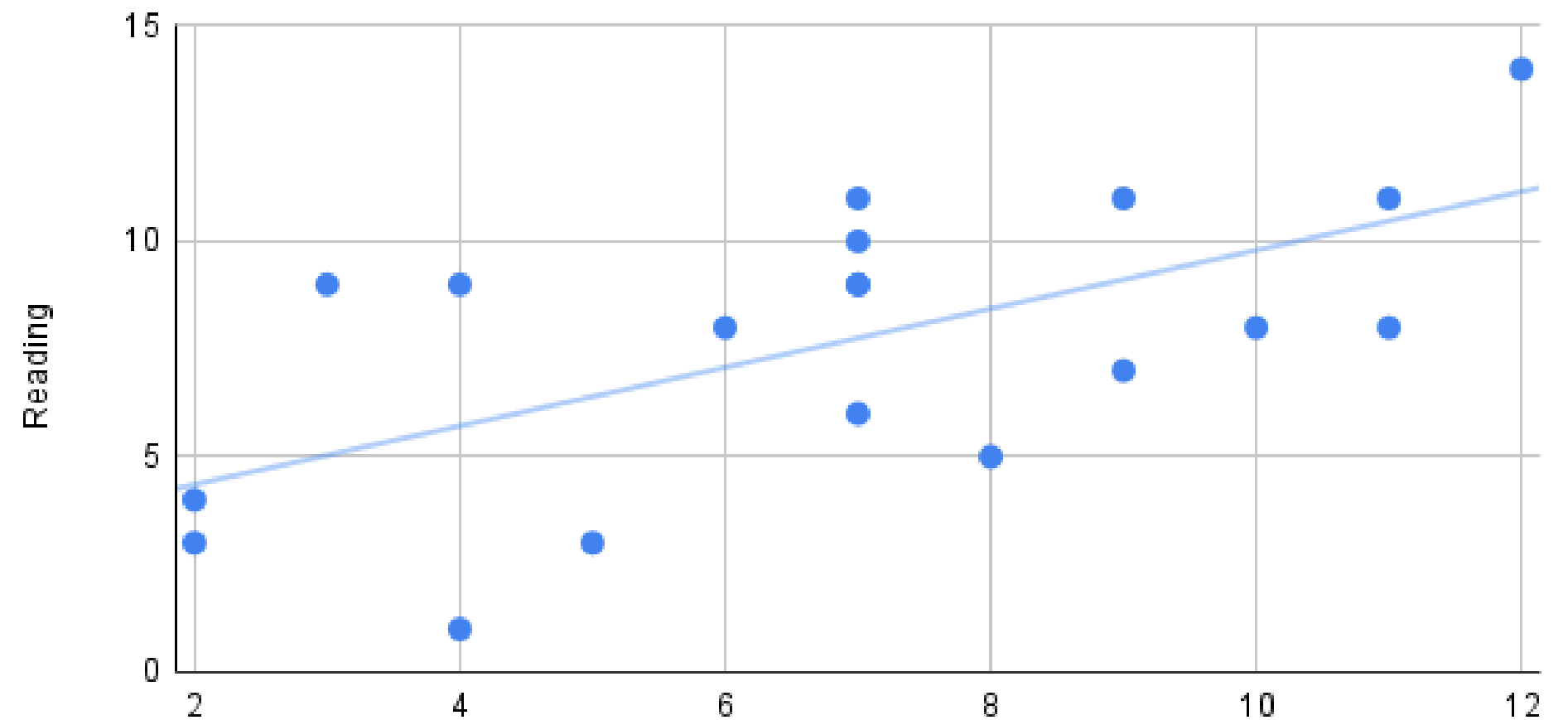
14 Reading Comprehension items

RESULTS

PEARSON'S CORRELATION TEST

Assumptions Checked

- Data is continuous
- Normally distributed ($p = .501$)
- Has linear relationship





¹ 3 Ms - Mode, median, midpoint

Descriptive Statistics

Critical Reading construct:

- Slightly negatively skewed
- Fairly platykurtic data
- 3 Ms¹ have closely tied
- Score range: 1-11

Analytical construct:

- Almost no skewness
- Relatively platykurtic data
- 3 Ms have same scores
- Score range: 1-13

Problem-Solving Skill	
N	19
Mean	6.89
Mode	7
Median	7
Min	2
Max	12
Midpoint	7
Range	10
Variance	9.09
SD	3.01
Skewness	-.06
Kurtosis	-.84

Reading Comprehension Skill	
N	19
Mean	7.68
Mode	9
Median	8
Min	1
Max	14
Midpoint	7.5
Range	13
Variance	11
SD	3.31
Skewness	-.33
Kurtosis	-.24

SD and Mean have closely tied scores between reading comprehension and problem-solving constructs

Item Analysis Tables

Cut-off score: 15/26 (60%)

NS + NNS

Most difficult item: item 20
(*IF* = 0.21)

Easiest item: item 7 (*IF* = 0.95)

NS

Most difficult item: items 8 & 20
(*IF* = 1.00)

Easiest item: items 5 & 7 (*IF* = 1.00)

NNS

Most difficult item: item 20
(*IF* = 0.00)

Easiest item: item 7 & 25 (*IF* = 0.88)

Appendix A. Native Speakers Item Analysis of Sentence Equivalence Items

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10	Item 11	Item 12
Test Taker 1	1	1	1	1	1	1	1	1	1	1	1	1
Test Taker 2	1	1	0	1	1	1	1	1	1	1	1	1
Test Taker 3	1	1	1	1	1	1	1	0	0	0	1	1
Test Taker 4	1	1	1	1	1	1	1	0	1	1	1	1
Test Taker 5	0	1	0	1	1	0	1	0	1	0	1	1
Test Taker 6	1	1	0	0	1	1	1	0	1	1	1	1
Test Taker 7	1	0	1	0	1	0	1	0	1	0	1	1
Test Taker 8	1	0	0	0	1	1	1	1	0	0	1	1
Test Taker 9	1	1	0	0	1	0	1	0	0	1	0	0
Test Taker 10	1	0	1	0	1	1	1	1	0	1	0	0
Test Taker 11	0	0	0	0	1	0	1	0	0	0	0	0
IF	0.82	0.64	0.45	0.45	1.00	0.64	1.00	0.36	0.55	0.55	0.73	0.73
ID	0.21	0.42	0.17	0.63	0.00	0.42	0.00	0.04	0.75	-0.17	1.00	1.00

Appendix B. Native Speakers Item Analysis of Reading Comprehension Items

	Item 13	Item 14	Item 15	Item 16	Item 17	Item 18	Item 19	Item 20	Item 21	Item 22	Item 23	Item 24	Item 25
Test Taker 1	1	1	1	1	1	1	1	1	1	1	1	1	1
Test Taker 2	0	1	1	1	1	0	1	0	1	0	0	1	1
Test Taker 3	1	0	0	1	1	0	1	1	1	1	1	1	1
Test Taker 4	1	1	1	0	0	1	1	0	1	0	0	0	1
Test Taker 5	0	1	1	1	1	1	1	0	1	1	1	0	1
Test Taker 6	0	1	0	0	1	0	0	0	1	1	1	0	1
Test Taker 7	1	1	1	0	1	1	1	0	1	1	1	1	0
Test Taker 8	0	1	0	1	0	0	1	1	1	1	1	1	1
Test Taker 9	0	0	1	0	1	1	1	1	1	1	0	1	0
Test Taker 10	1	0	0	0	1	0	0	0	1	0	0	1	0
Test Taker 11	0	1	0	0	0	0	1	0	0	0	0	1	0

Appendix C. Non-Native Speakers Item Analysis of Sentence Equivalence Items

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10	Item 11	Item 12	TOTAL
Test Taker 12	1	1	1	1	1	1	1	0	1	1	1	1	11
Test Taker 13	1	1	0	0	1	1	1	1	0	1	0	0	7
Test Taker 14	0	0	0	1	1	0	1	0	0	0	0	1	4
Test Taker 15	1	1	1	1	0	0	1	0	1	0	0	1	7
Test Taker 16	0	0	0	0	1	0	1	0	0	0	1	0	3
Test Taker 17	0	0	0	1	1	1	1	0	0	1	0	0	5
Test Taker 18	0	0	0	0	0	0	0	0	0	0	1	1	2
Test Taker 19	0	0	0	0	0	1	1	0	1	0	0	1	4
IF	0.38	0.38	0.25	0.50	0.63	0.50	0.88	0.13	0.38	0.38	0.38	0.63	
ID	0.83	0.83	0.33	0.00	0.50	0.67	0.17	0.50	0.17	0.83	0.17	-0.17	

Appendix D. Non-Native Speakers Item Analysis of Reading Comprehension Items

	Item 13	Item 14	Item 15	Item 16	Item 17	Item 18	Item 19	Item 20	Item 21	Item 22	Item 23	Item 24	Item 25	Item 26	TOTAL
Test Taker 12	1	1	0	0	1	1	1	0	1	1	1	1	1	1	11
Test Taker 13	1	0	0	0	1	0	1	0	1	1	1	1	1	1	9
Test Taker 14	0	1	1	0	1	1	0	0	1	1	1	0	1	1	9
Test Taker 15	0	1	0	1	0	0	0	0	0	1	1	1	1	0	6
Test Taker 16	0	1	1	0	1	1	1	0	1	1	0	1	1	0	9
Test Taker 17	0	0	0	1	0	0	0	0	0	0	0	1	1	0	3
Test Taker 18	0	0	1	1	0	0	0	0	0	1	0	0	1	0	4
Test Taker 19	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1
IF	0.38	0.50	0.38	0.38	0.50	0.38	0.38	0.00	0.50	0.75	0.50	0.63	0.88	0.38	
ID	0.83	0.00	-0.50	-0.50	0.67	0.17	0.83	0.00	0.67	0.33	0.67	0.50	0.17	0.83	

Analytical Construct

Appendix A. Native Speakers Item Analysis of Sentence Equivalence Items

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10	Item 11	Item 12	TOTAL
Test Taker 1	1	1	1	1	1	1	1	1	1	1	1	1	12
Test Taker 2	1	1	0	1	1	1	1	1	1	1	1	1	11
Test Taker 3	1	1	1	1	1	1	1	0	0	0	1	1	9
Test Taker 4	1	1	1	1	1	1	1	0	1	1	1	0	10
Test Taker 5	0	1	0	1	1	0	1	0	1	0	1	1	7
Test Taker 6	1	1	0	0	1	1	1	0	1	1	1	1	9
Test Taker 7	1	0	1	0	1	0	1	0	1	0	1	1	7
Test Taker 8	1	0	0	0	1	1	1	1	0	0	1	1	7
Test Taker 9	1	1	0	0	1	0	1	0	0	1	0	1	6
Test Taker 10	1	0	1	0	1	1	1	1	0	1	0	1	8
Test Taker 11	0	0	0	0	1	0	1	0	0	0	0	0	2
IF	0.82	0.64	0.45	0.45	1.00	0.64	1.00	0.36	0.55	0.55	0.73	0.82	
ID	0.21	0.42	0.17	0.63	0.00	0.42	0.00	0.04	0.75	-0.17	1.00	0.21	

Appendix C. Non-Native Speakers Item Analysis of Sentence Equivalence Items

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10	Item 11	Item 12	TOTAL
Test Taker 12	1	1	1	1	1	1	1	0	1	1	1	1	11
Test Taker 13	1	1	0	0	1	1	1	1	0	1	0	0	7
Test Taker 14	0	0	0	1	1	0	1	0	0	0	0	1	4
Test Taker 15	1	1	1	1	0	0	1	0	1	0	0	1	7
Test Taker 16	0	0	0	0	1	0	1	0	0	0	1	0	3
Test Taker 17	0	0	0	1	1	1	1	0	0	1	0	0	5
Test Taker 18	0	0	0	0	0	0	0	0	0	0	1	1	2
Test Taker 19	0	0	0	0	0	1	1	0	1	0	0	1	4
IF	0.38	0.38	0.25	0.50	0.63	0.50	0.88	0.13	0.38	0.38	0.38	0.63	
ID	0.83	0.83	0.33	0.00	0.50	0.67	0.17	0.50	0.17	0.83	0.17	-0.17	

Critical Reading Construct

Appendix B. Native Speakers Item Analysis of Reading Comprehension Items

	Item 13	Item 14	Item 15	Item 16	Item 17	Item 18	Item 19	Item 20	Item 21	Item 22	Item 23	Item 24	Item 25	Item 26	TOTAL
Test Taker 1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	14
Test Taker 2	0	1	1	1	1	0	1	0	1	0	0	1	1	0	8
Test Taker 3	1	0	0	1	1	0	1	1	1	1	1	1	1	1	11
Test Taker 4	1	1	1	0	0	1	1	0	1	0	0	0	1	1	8
Test Taker 5	0	1	1	1	1	1	1	0	1	1	1	0	1	1	11
Test Taker 6	0	1	0	0	1	0	0	0	1	1	1	0	1	1	7
Test Taker 7	1	1	1	0	1	1	1	0	1	1	1	1	0	0	10
Test Taker 8	0	1	0	1	0	0	1	1	1	1	1	1	1	0	9
Test Taker 9	0	0	1	0	1	1	1	1	1	0	1	1	0	0	8
Test Taker 10	1	0	0	0	1	0	0	0	1	0	0	1	0	1	5
Test Taker 11	0	1	0	0	0	0	1	0	0	0	0	1	0	0	3
IF	0.45	0.73	0.55	0.45	0.73	0.45	0.82	0.36	0.91	0.55	0.64	0.73	0.64	0.55	
ID	0.17	0.54	0.29	0.63	0.08	0.04	0.21	0.04	0.33	0.75	0.42	-0.38	0.88	0.29	

Appendix D. Non-Native Speakers Item Analysis of Reading Comprehension Items

	Item 13	Item 14	Item 15	Item 16	Item 17	Item 18	Item 19	Item 20	Item 21	Item 22	Item 23	Item 24	Item 25	Item 26	TOTAL
Test Taker 12	1	1	0	0	1	1	1	0	1	1	1	1	1	1	11
Test Taker 13	1	0	0	0	1	0	1	0	1	1	1	1	1	1	9
Test Taker 14	0	1	1	0	1	1	0	0	1	1	1	0	1	1	9
Test Taker 15	0	1	0	1	0	0	0	0	0	1	1	1	1	0	6
Test Taker 16	0	1	1	0	1	1	1	0	1	1	0	1	1	0	9
Test Taker 17	0	0	0	1	0	0	0	0	0	0	0	1	1	0	3
Test Taker 18	0	0	1	1	0	0	0	0	0	1	0	0	1	0	4
Test Taker 19	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1
IF	0.38	0.50	0.38	0.38	0.50	0.38	0.38	0.00	0.50	0.75	0.50	0.63	0.88	0.38	
ID	0.83	0.00	-0.50	-0.50	0.67	0.17	0.83	0.00	0.67	0.33	0.67	0.50	0.17	0.83	

TOTAL SCORES

of all participants

Test Taker 1	25
Test Taker 2	20
Test Taker 3	20
Test Taker 4	18
Test Taker 5	18
Test Taker 6	17
Test Taker 7	17
Test Taker 8	16
Test Taker 12	22
Test Taker 9	14
Test Taker 10	14
Test Taker 11	5
Test Taker 14	13
Test Taker 15	13
Test Taker 16	12
Test Taker 17	9
Test Taker 18	6
Test Taker 19	5

Easiest Item: Item 7

International financial issues are typically _____ by the United States media because they are too technical to make snappy headlines and too inaccessible to people who lack a background in economics.

- ☐ neglected
- ☐ slighted
- ☐ overrated
- ☐ hidden
- ☐ criticized
- ☐ repudiated

Easiest Item: Item 7

International financial issues are typically _____ by the United States media because they are too technical to make snappy headlines and too inaccessible to people who lack a background in economics.

☒ neglected

☒ slighted

☐ overrated

☐ hidden

☐ criticized

☐ repudiated

In Raisin in the Sun, Lorraine Hansberry does not reject integration or the economic and moral promise of the American dream; rather, she remains loyal to this dream while looking, realistically, at its incomplete realization. Once we recognize this dual vision, we can accept the play's ironic nuances as deliberate social commentaries by Hansberry rather than as the "unintentional" irony that Bigsby attributes to the work. **Indeed, a curiously persistent refusal to credit Hansberry with a capacity for intentional irony has led some critics to interpret the play's thematic conflicts as mere confusion, contradiction, or eclecticism.** Isaacs, for example, cannot easily reconcile Hansberry's intense concern for her race with her ideal of human reconciliation. But the play's concern for racial justice and human solidarity as complementary goals is evident in Du Bois's famous, well-considered vision of a world coexisting with human unity, of an internationalism that also accommodates individual roles.

Most Difficult Item: Item 20

Select and indicate the best answer from among the five answer choices:

The author's primary purpose in the passage is to

- ☐ explain some critics' refusal to consider *Raisin in the Sun* a deliberately ironic play
- ☐ suggest that ironic nuances ally *Raisin in the Sun* with Du Bois's and Fanon's writings
- ☐ analyze the fundamental dramatic conflicts in *Raisin in the Sun*
- ☐ emphasize the inclusion of contradictory elements in *Raisin in the Sun*
- ☐ affirm the thematic coherence underlying *Raisin in the Sun*

In Raisin in the Sun, Lorraine Hansberry does not reject integration or the economic and moral promise of the American dream; rather, she remains loyal to this dream while looking, realistically, at its incomplete realization. Once we recognize this dual vision, we can accept the play's ironic nuances as deliberate social commentaries by Hansberry rather than as the "unintentional" irony that Bigsby attributes to the work. Indeed, a curiously persistent refusal to credit Hansberry with a capacity for intentional irony has led some critics to interpret the play's thematic conflicts as mere confusion, contradiction, or eclecticism. Isaacs, for example, cannot easily reconcile Hansberry's intense concern for her race with her ideal of human reconciliation. But the play's complex view of Black self-esteem and human solidarity as compatible is no more "contradictory" than Du Bois's famous, well-considered ideal of ethnic self-awareness coexisting with human unity, or Fanon's emphasis on an ideal internationalism that also accommodates national identities and roles.

Most Difficult Item: Item 20

te the best answer from among the five answer choices:

ary purpose in the passage is to

- ☐ explain some critics' refusal to consider Raisin in the Sun a deliberately ironic play
- ☐ suggest that ironic nuances ally Raisin in the Sun with Du Bois's and Fanon's writings
- ☐ analyze the fundamental dramatic conflicts in Raisin in the Sun
- ☐ emphasize the inclusion of contradictory elements in Raisin in the Sun
- ☒ affirm the thematic coherence underlying Raisin in the Sun

Item Discrimination

1. Small sample size:

- NS: 8 proficient, 4 less proficient
- NNS: 2 proficient, 6 less proficient

2. Negative scores: less proficient groups got the items right, and more proficient groups did not get the item right.

3. Out of 11 NS test takers, only one proficient (69.23%) test taker in the proficient group got both of two most problematic items wrong.
($ID_1 = -.17$; $ID_2 = -.38$)

4. Out of 9 NNS test takers, only one proficient (61.93%) test taker scored all zeros in three most problematic items.
($ID_1 = -.17$; $ID_2 = -.50$; $ID_3 = -.50$)



Correlation

- Relationships between critical reading and analytical constructs
- Significant result ($p = 0.005$)
- Positive correlation ($r = .61$, $r^2 = .37$)
- Reliability coefficient: test is 61% consistent for both variables
- Coefficient of determination: 37% of shared variance between the two constructs.

Reliability

- Reliability of test scores between NS and NNS
- Overall reliability of over 0.8 ($\alpha = .85$), interpreted as good reliability (George & Mallery, 2016).
- $\alpha = .82$ for NS
- $\alpha = .80$ for NNS
- Very slight difference in reliability levels between two groups
- Test is reliable for both groups as these reliability values are fairly high
- 39% measurement error: if the assessment was taken again, the test takers might possibly have earned total scores that are different from their current scores

DISCUSSION

Research Questions Response

1. What is the relationship between critical reading and analytical skills in GRE-V?

- Pearson's test showed a positive correlation between two constructs; 37% of shared variance -> two constructs are related

2. To what extent are the test scores of English native speakers and English non-native speakers different?

- They had some easy and difficult items in common, even when separated; normal distribution for both constructs with some skewness and flattened data; NNS sample size was smaller -> item discrimination issues; no participant had a maximum score. There was only one person who score 25/26

3. How consistent are the test scores for NS and NNS groups?

- High reliability scores for both groups when separated and combined



IMPLICATIONS

EDUCATIONAL IMPLICATIONS

1. Can be used for both NS & NNS test takers
2. Reliable for NNS, even though the assumptions might be that NNS might perform much lower in score rates than NS (the scores were lower in NNS group, but it might also be related to a small sample size)
3. There was no deviation in normal distribution -> significant results
4. Difficult for both groups: a few PhD students had higher scores than any participants
5. Test takers need to prepare and take several tests beforehand to be able to perform highly.



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THANK YOU!

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